Pre-match Tennis Winner Prediction with Machine Learning

UE24CS352A Machine Learning, Mini-Project · Harish Ramesh Kumar (PES1UG24AM111) · Jayant Mahishi (PES1UG24AM122) · github.com/harish-rameshkumar/tennis-match-prediction

1 PROBLEM STATEMENT

Predict the winner of an ATP men's singles match *before it starts*, using only information available at that point. We reproduce A. K. Amudan, *Tennis Match Prediction using Machine Learning* (Stanford CS229). The paper reports 71% accuracy with logistic regression, against a 65.5% baseline of always picking the higher-ranked player.

2 DATASET

Source. Jeff Sackmann's `tennis_atp` match data (CC BY-NC-SA 4.0). The original GitHub repository is no longer available, so we download it from the archival mirror `Aneeshers/tennis-sackmann-archive`, pinned to one commit and checked against SHA-256 checksums: 43 CSVs (40 MB) covering the ATP main tour 1997–2024, plus Challenger and qualifying matches 2010–2024.

Cleaning. We drop rows with a missing player, date or rank, as well as walkovers and score-less rows. Retirements are kept. This leaves 196,780 of 205,168 matches.

Splits. Train on every match before 2019 (131,562 matches, including Challenger, qualifying and Davis Cup). Test on the 2,597 ATP main-tour matches of 2019 (Grand Slams, Masters, ATP 250/500, Finals). For a robustness check, train on matches up to 2022 and test on 5,464 main-tour matches from 2023–24.

3 APPROACH

- **Mirrored rows.** Each match becomes two rows: (A = winner, label 1) and (A = loser, label 0), so a model cannot learn which slot the winner is in. At test time the two predictions are combined into one per match, P(A wins) = ½·(p(A,B) + 1 − p(B,A)), and accuracy is counted per match.
- **65 features.** The paper's player attributes (rank, ranking points, height, hand, age, kept as raw values), with missing-value flags and medians computed from the training years only. Then **47 history features**, mostly as A − B differences: career, last-5/10/15/25, last-30-day, surface and same-tournament records; head-to-head (overall and on this surface) and win rate against common opponents; career serve averages; overall and surface Elo.
- **No leakage.** A match's history is every tournament with an earlier start date, plus earlier rounds of the same tournament. Tests rebuild every feature from explicitly filtered data for 62 sampled matches, and check that flipping a future result changes no earlier feature. Two deliberately leaky versions of the pipeline were both caught.
- **Models** (scikit-learn): rank baseline, logistic regression, LinearSVC (calibrated), random forest, histogram gradient boosting.

4 IMPLEMENTATION OVERVIEW

The code is a Python 3.13 package, `src/tennis`: `load` → `clean` → `features` (one chronological pass with per-player state) → `dataset` (mirrored `pairs.parquet`, `get_train_test`) → `models` / `evaluate`. For a new match, `inference` builds the rows with the *same* feature code (20 tested past matches reproduce their stored training rows exactly), and `predict` serves a Streamlit app and a command-line fallback. 40 automated tests pass. On a MacBook Air, the data and features build in under a minute, and training and evaluating all five models takes about 2 minutes.

5 RESULTS

Model	2019 test accuracy	2019 train accuracy	2019 test log-loss	2019 test Brier	2023–24 test accuracy
Rank baseline (higher rank wins)	61.3%	65.0%	—	0.387	63.5%
Logistic regression	65.7%	67.8%	0.613	0.213	66.0%
LinearSVC (calibrated)	**66.2%**	67.8%	0.612	0.212	**66.1%**
Random forest	66.1%	85.1%	0.611	0.213	65.8%
Gradient boosting (HGB)	66.1%	68.4%	**0.610**	**0.212**	**66.1%**

Per-match accuracy on 2,597 matches (2019) and 5,464 matches (2023–24). The baseline makes hard 0/1 predictions, so it has no log-loss.

6 CONCLUSIONS

- **All four learned models land within about half a point of each other** (65.7–66.2% on 2019, 65.8–66.1% on 2023–24). The features, not the model type, set the ceiling.
- **Random forest overfits**: 85.1% on its training data against 66.1% on test, with no gain over the simpler models.
- **We beat the rank baseline but do not reproduce the paper's 71%.** Our margin over the baseline is 4.4–4.9 points on 2019 (2.3–2.6 on 2023–24), against the paper's 5.5.
- **Why the baseline differs.** 2019 has the lowest rank baseline (61.3%) of any season from 1997 to 2024; 2015–2018 give 67.9%, 67.2%, 65.0% and 63.6%. This explains the gap to the paper's 65.5% baseline, so we compare against our own baseline.

Who built what. Harish: data pipeline, cleaning, the 47 history features and their leakage tests, the inference layer, the EDA notebook, README and command-line demo. **Jayant:** the models, evaluation and ablation, `predict_match`, the Streamlit app and the results notebook.